Project Dashboard - Comprehensive Project Tracking System

M. Zaeem | Electronics & Embedded Systems Engineer | BSc Electrical Engineering | MSc Autonomous Systems

1. Project Basic Information

Project Name:	Solar-Powered Automated Gong with IoT Data Logging		
Client Name:	Hierro Rebby		
Start Date:	11-Sep-25	**Expected End Date:**	30-Nov-25
Pricing Model:	**Fixed Price** ☑ **Hourly Rate** ☐		
Hourly Rate Per Hour ($):	$15.00	**Total Project Budget:**	
Payment by Milestone:	☑		

2. Project Requirements

Project Requirements

The project involves designing and building a self-sustaining, solar-powered system that automatically strikes a gong at high noon, triggered by a light sensor. The system must be future-proofed for a Phase 3 upgrade to include audio recording and cellular IoT data transmission.

Core Requirements (Phase 1):

Microcontroller: To be powered solely by solar energy. Must be capable of handling Phase 2 requirements (e.g., Raspberry Pi Pico, ESP32).
Actuation: A solenoid (or servo) controlled via a physical relay to strike the gong with a strong, audible sound.
Sensing: An LDR (Light Dependent Resistor) to detect high noon.
Power System: A complete off-grid power system including:
-Solar PV Panel
-Charge Controller
-Rechargeable Battery Pack (Li-ion recommended)
-Buck Converter to provide stable 5V/3.3V to the MCU and sensors.
Hardware Sourcing: Components must be sourced from reliable, non-Amazon suppliers (e.g., AliExpress, for a target budget of under £200, prioritizing value for potential duplication.

Future-Proofing for Phase 2:

The hardware specification must accommodate the future addition of:
A microphone (Electric or MEMS, 1.5V-3.6V) to record the gong strike.
A Cellular IoT module with built-in GPS.
Increased energy storage and power management to handle the additional load.

3. Project Milestones

	Start Date:		
	Task Requirements	**Description**	**Status**
Milestone 1	Finalize System Architecture	in process	Complete
	Finalising Components and Create Shopping List		Complete
	Hardware Assembly		Pending
	Firmware Development		Pending
	Real-World Unit Testing		Pending
	Delivery to Client		Pending
	Task Requirements	**Description**	**Status**
Milestone 2			
	Task Requirements	**Description**	**Status**
Milestone 3			

4. Key Deliverables

		Additional Requirements	Project End Goal and Approval Criteria
Sensor Selection	☑		All hardware components functioning to specification Software meets All requirements with no critical bugs Client signs off on final deliverables
Firmware Development	☑		
Schematic Drawing	☐		
PCB Layout	☐		
Gerber Files	☐		
Bill of Materials (BOM)	☑ in process	Clarifications + Value Engineering	
PCB Fabrication Ordering	☐		
PCB Assembly (soldering components)	☐		
Board Testing	☐		
3D Enclosure Design	☐		

5. Additional Expenses :

Item / Description	Category	Notes	Cost per Unit ($)	Units	Total ($)
PCB Fabrication (JLCPCB)	Fabrication				$0.00
Customs VAT	Customs/Tax				$0.00
Enclosure Printing	Outsourcing				$0.00
PCB Assembly	Assembly Service				$0.00
Components	Components LCSC				$0.00
Unit Testing Labor	Testing				$0.00
Delivery to Client	Logistics				$0.00
Others					$0.00
				Total Cost:	**$0.00**

6. Payment Section

Fixed Priced Milestone Payments

Task / Milestone	Fixed Amount ($)	Description	Payment Date	Paid ($)	Remaining ($)
Milestone 1	$235.00			$235.00	$0.00
Milestone 2					$0.00
Milestone 3					$0.00
Additional Expenses	$0.00				$0.00
Total Amount:				**$235.00**	**$0.00**

Thank you for choosing to work with me. I truly appreciate your trust, and I'll do my best to bring your idea to life with precision, dedication, and transparency. This spreadsheet has been created to provide a clear overview of project scope, progress, and costs, ensuring full visibility at every step.

Please Note:

The quotation provided does not include external expenses such as:
- Component sourcing (AliExpress, Digi-Key, etc.)
- PCB fabrication or assembly services (e.g., JLCPCB, PCBWay)
- Import duties, VAT, or customs clearance fees
- Packaging or shipping of prototypes
- Specialized tooling or any outsourced service costs
- These costs will be discussed and billed separately if required during the project.

Looking forward to building something great together from concept to completion, I'm here as your technical partner.

Warm regards,
M. Zaeem
Electronics & Embedded Systems Engineer